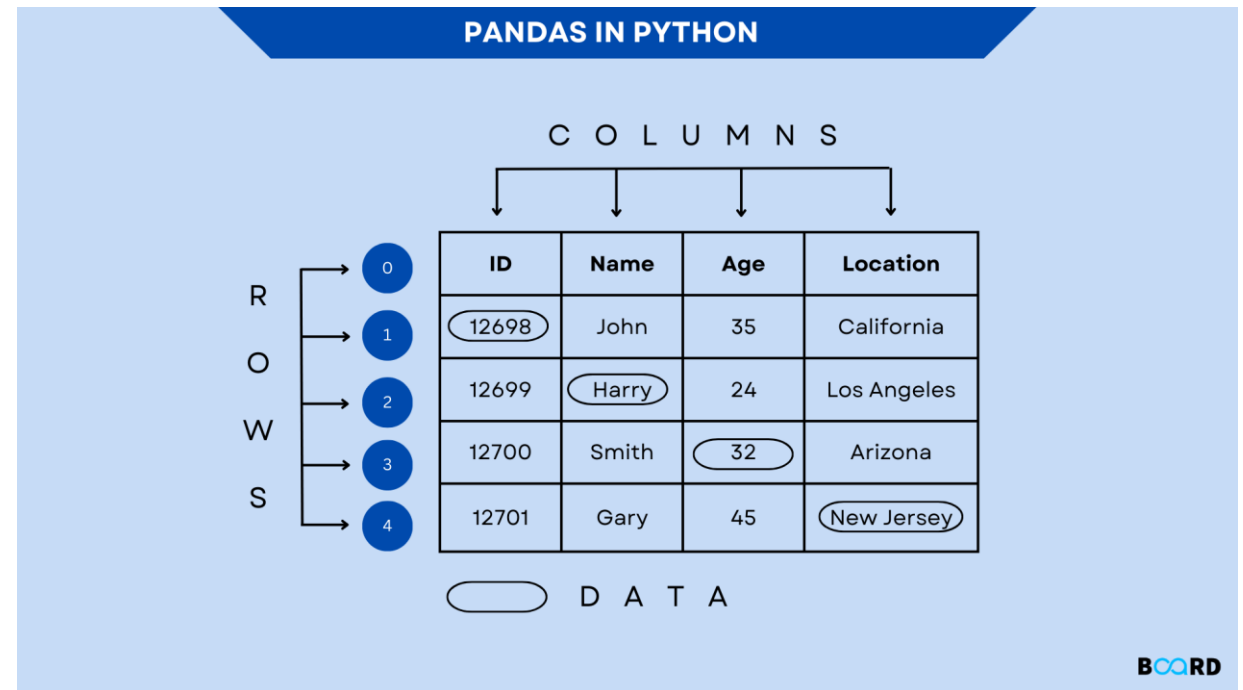


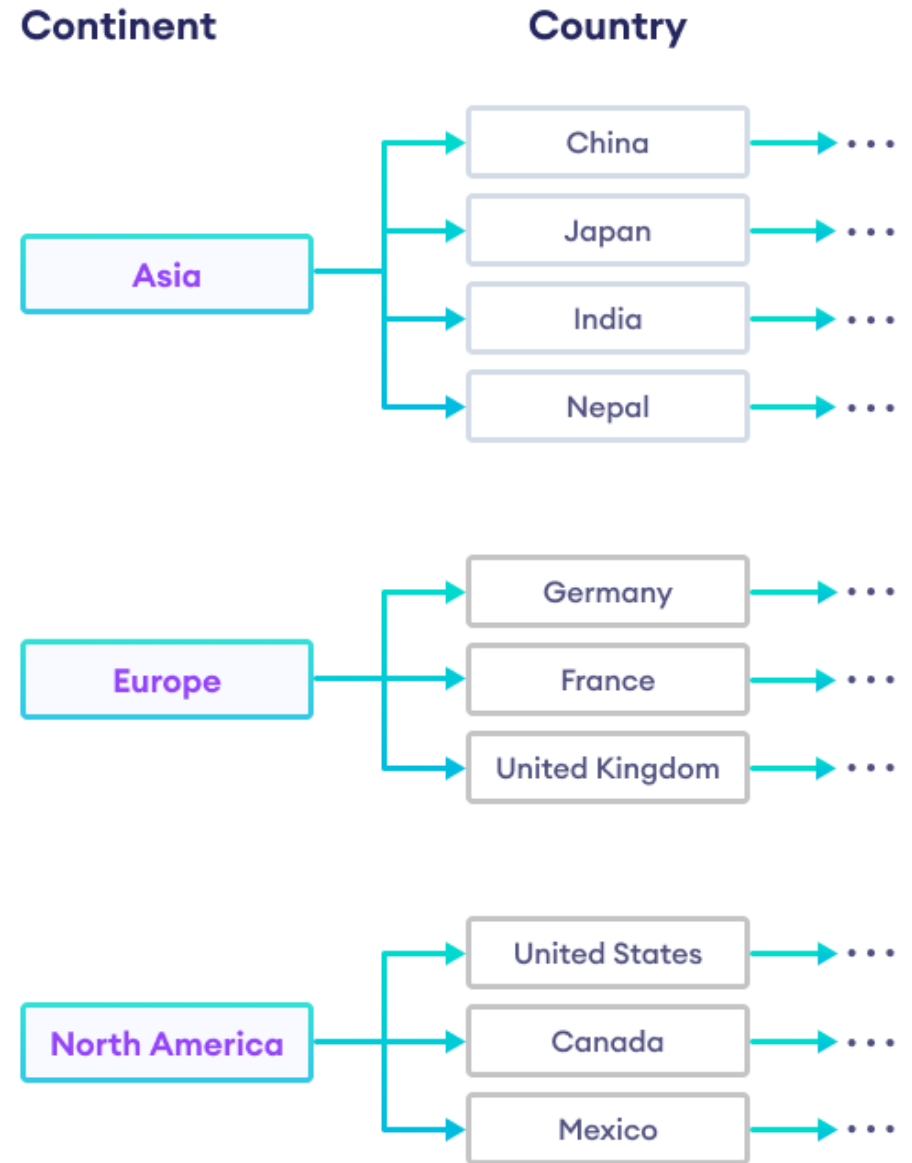
The Anatomy of Pandas



- **Series (1D):** Think of this as a single column in an Excel sheet. It is an array of data with an associated label called an *index*.
- **DataFrames (2D):** The primary tool for data analysis. It is a collection of Series sharing the same index. It represents the standard "Table" format used in almost every industry.
- **The Power of Labels:** Unlike NumPy, where you must remember index [0, 1], Pandas allows you to access data by names like 'Price' or 'Date'

Multi-Indexing (Hierarchical Data)

- **Beyond 2D:** What if your data has layers? (e.g., Data for multiple **Cities**, collected over multiple **Years**, across different **Months**).
- **Organization:** Multi-indexing allows you to store and query high-dimensional data in a 2D table format.
- **Regional Use Case:** This is essential for "Regional Analytics," such as comparing agricultural yields across different districts in KPK over a 10-year span.



Categorical Data: The Invisible Speed Boost

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 100 entries, 0 to 99
```

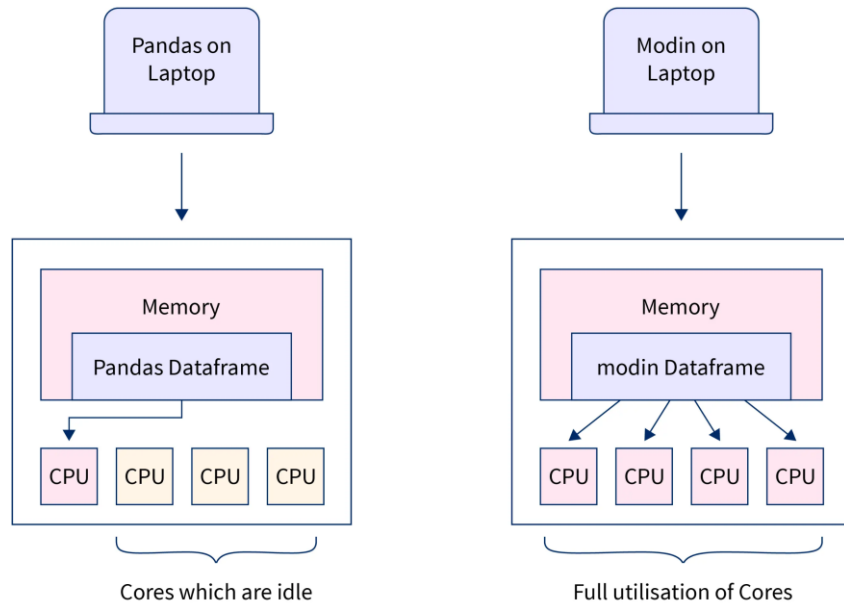
```
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	age	100 non-null	int64
1	coding_exp	100 non-null	object
2	title	100 non-null	object
3	fav_myth_creature	100 non-null	object
4	fav_3_letters	100 non-null	object
5	secret_group	100 non-null	int64

```
dtypes: int64(2), object(4)
```

```
memory usage: 4.8+ KB
```

- **The Problem:** Storing the word "Peshawar" 1 million times as a string is a waste of memory.
- **The Solution:** The category data type stores the unique string once and uses small integers to represent the repetitions.
- **Impact:** This can reduce file sizes by up to 90% and speed up grouping operations significantly.



SCALER
Topics

Memory Optimization & Scaling

- **The Laptop Limit:** Standard laptops crash when a dataset exceeds available RAM.
- **Downcasting:** Converting float64 to float32 or int64 to int16.
- **Efficient Loading:** Only loading the columns you need. We teach these techniques so you can process millions of rows on standard hardware.